

FastAPI Dependency Injection Issue SOP

Objective

Resolve dependency injection failures where FastAPI dependencies are not resolving correctly or are returning invalid objects.

Scope

Applicable to FastAPI-based services, backend APIs, API gateways, DevOps teams, QA teams, and platform teams supporting production, staging, or synthetic incident workflows.

Pre-requisites

1. Access to FastAPI application logs, deployment logs, and API gateway or ingress logs.
2. Access to monitoring dashboards such as Grafana, Prometheus, APM, or cloud monitoring.
3. Permission to view application configuration, environment variables, and deployment status.
4. API testing tool such as curl, Postman, Swagger UI, or automated smoke test scripts.
5. Escalation contact for backend, DevOps, database, and security teams.

Incident Metadata

Field	Value
Ticket ID	T026
Issue	FastAPI dependency injection issue
Category	API
Service	FastAPI
Severity	Low
Status	Resolved
Description	Dependencies not resolving correctly
Expected Resolution	Check dependency function definitions

Related Log Evidence

No direct matching log entry was present in the provided logs CSV. Use the ticket description, application logs, API gateway logs, and monitoring dashboards to validate this incident in runtime.

Procedure

1. Identify the route and dependency function producing the failure.
2. Review stack trace for dependency resolution, Depends usage, or callable signature errors.
3. Validate dependency function parameters, return type, and required request context.
4. Check nested dependencies for missing configuration, database sessions, or auth context.
5. Confirm sync versus async dependency behavior is correct.
6. Add unit tests for dependency success and failure paths.
7. Fix dependency definition or route injection and redeploy.
8. Verify route behavior using Swagger UI or curl.

Common Commands

```
curl -i -X GET http://<host>/health
curl -i -X POST http://<host>/<endpoint> -H "Content-Type: application/json" -d @payload.json
```

```
kubectll get pods -n <namespace> | grep fastapi
kubectll logs <fastapi-pod> -n <namespace> --tail=100
kubectll describe pod <fastapi-pod> -n <namespace>
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Dependency not resolving	Incorrect Depends definition	Fix callable and route signature
DB session missing	Session dependency misconfigured	Correct session lifecycle
Auth context missing	Dependency order issue	Validate auth dependency chain
Async error	Sync/async mismatch	Use correct async function pattern

Best Practices

1. Keep dependency functions small and testable.
2. Use typed return values for shared dependencies.
3. Avoid hidden global state in dependencies.
4. Test nested dependency chains.

References

FastAPI Docs | Internal Logs CSV | Tickets CSV | Monitoring Dashboards | API Gateway Logs | Backend Runbook